

Deep Learning: Theory and Practice

Advanced Topics in Deep Learning

14-5-2020

Supervised

$$\underline{x} \xrightarrow{NN} y, t$$

$$\downarrow L(y, t)$$

Unsupervised

$$\underline{x}$$

→ Generate new data

→ Can I combine unsupervised with supervised small data

Deep Unsupervised Learning

Active learning

Resources to label "n"

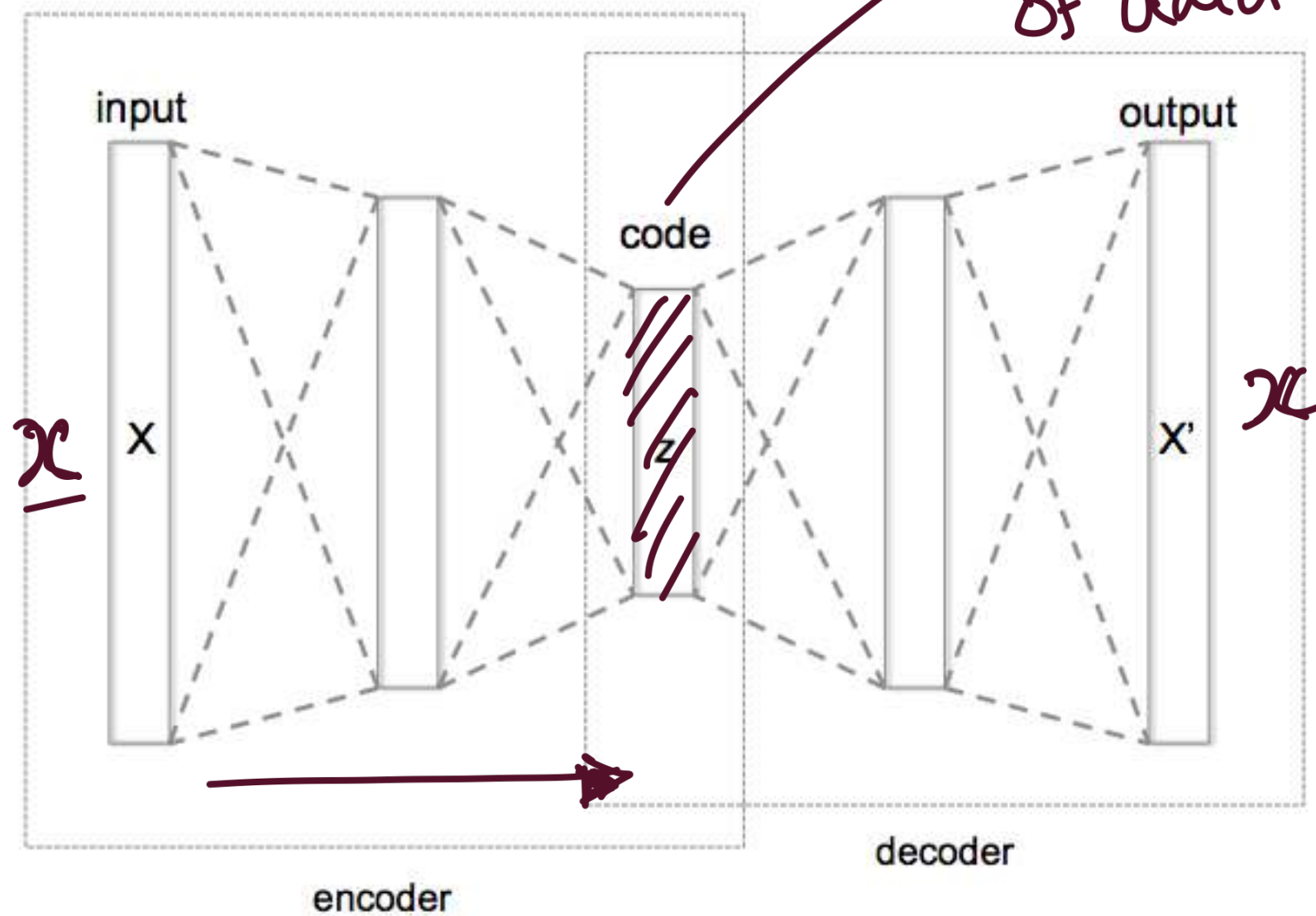
$$\{x_i\}_1^N$$

$$\text{Test data } \{x_i\}_1^M$$

$$(n) \ll (N)$$

$$N = 100,000$$
$$n = 1,000$$

Autoencoders

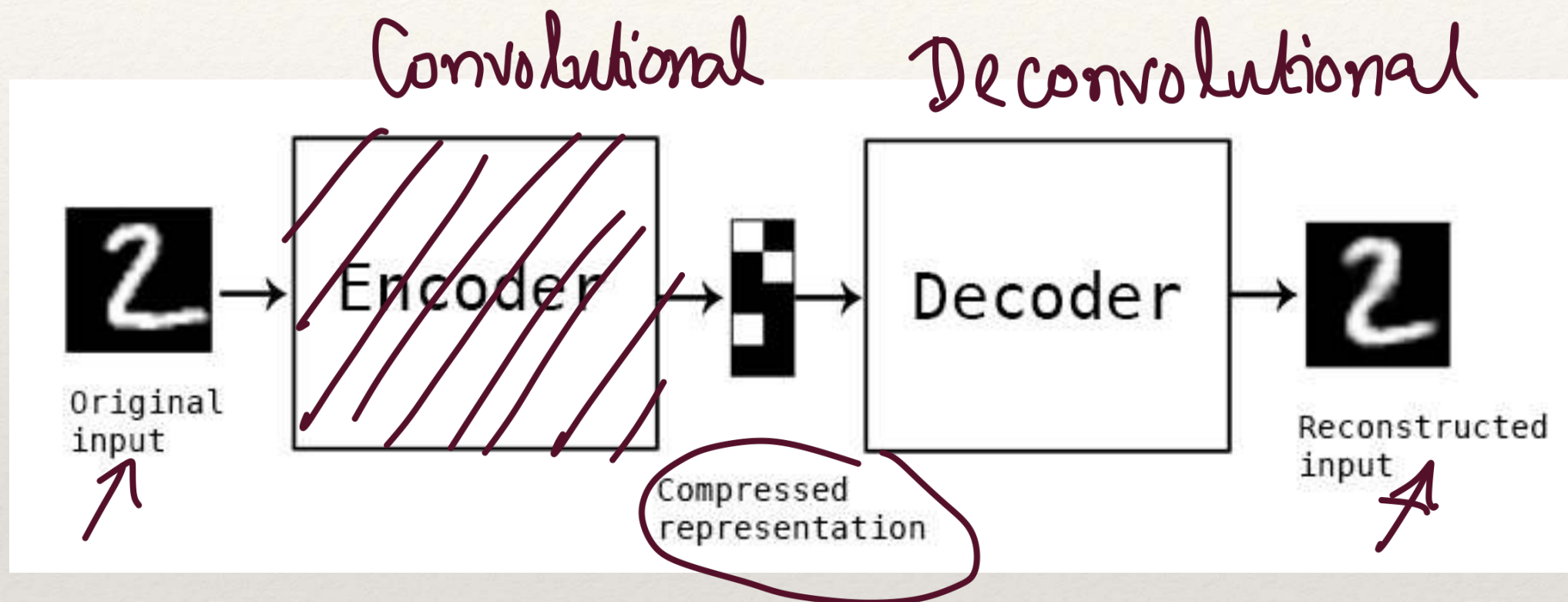


$$\mathcal{X} = \frac{1}{N} \sum \mathcal{X}$$

Training loss

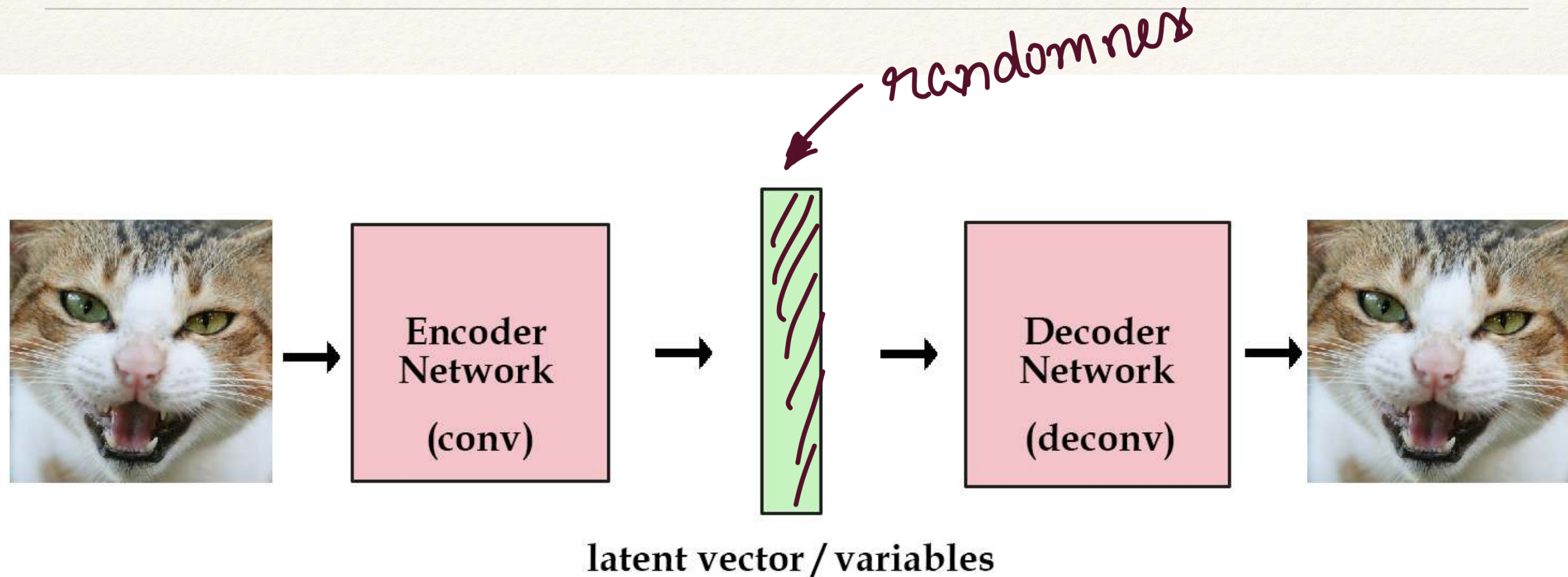
**Avoid
Identity
Mapping**

Autoencoders



Encoder $\rightarrow$ LSTM, sequence to vector
Decoder $\rightarrow$ LSTMs. vector to sequence

Convolutional Autoencoders

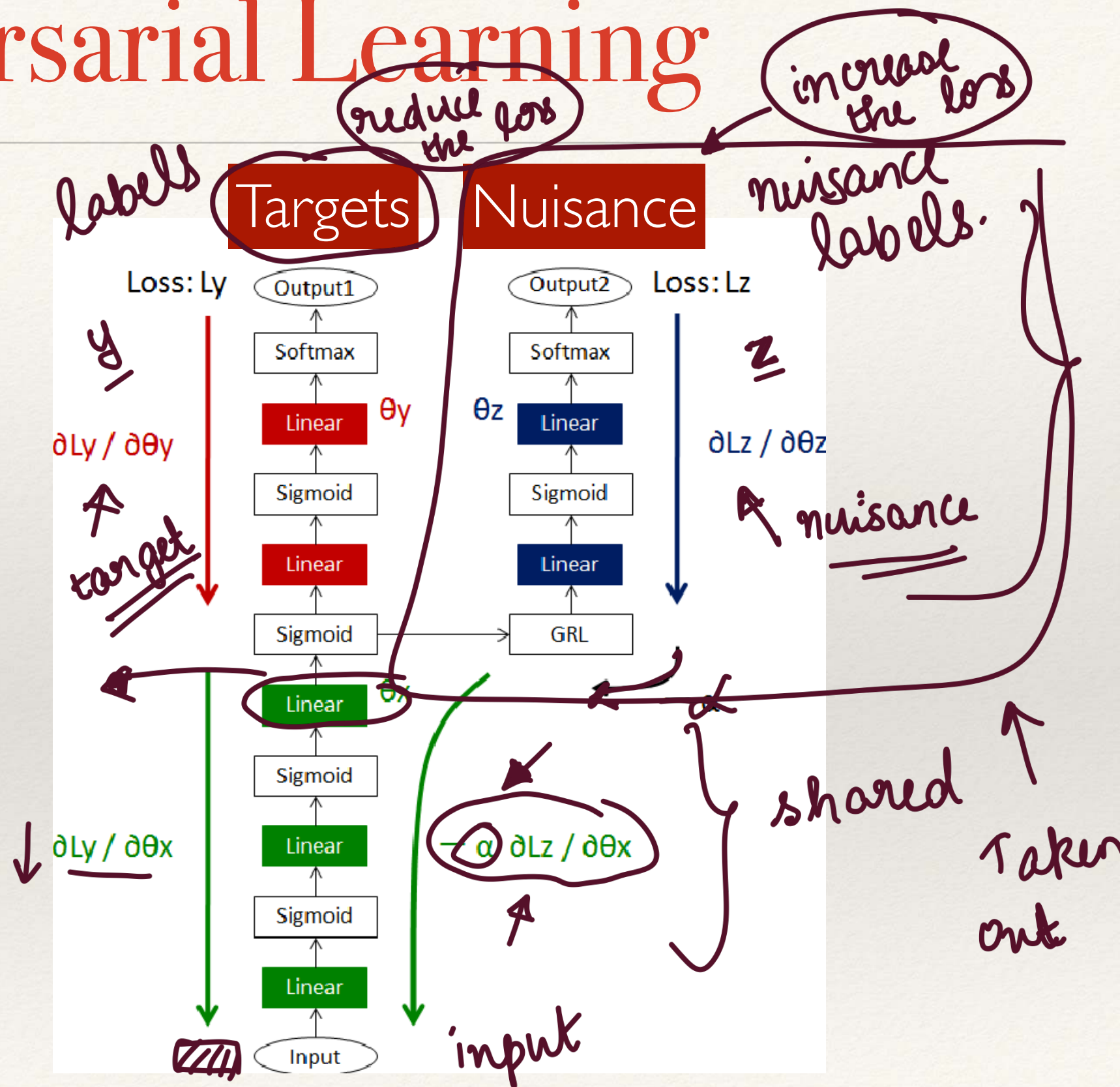


The latent vectors can form deep features for other supervised tasks.

Supervised Adversarial Learning

- The target can be learned
 - using original gradient.
 - domain adversarial gradient.
- Model will learn to be domain invariant

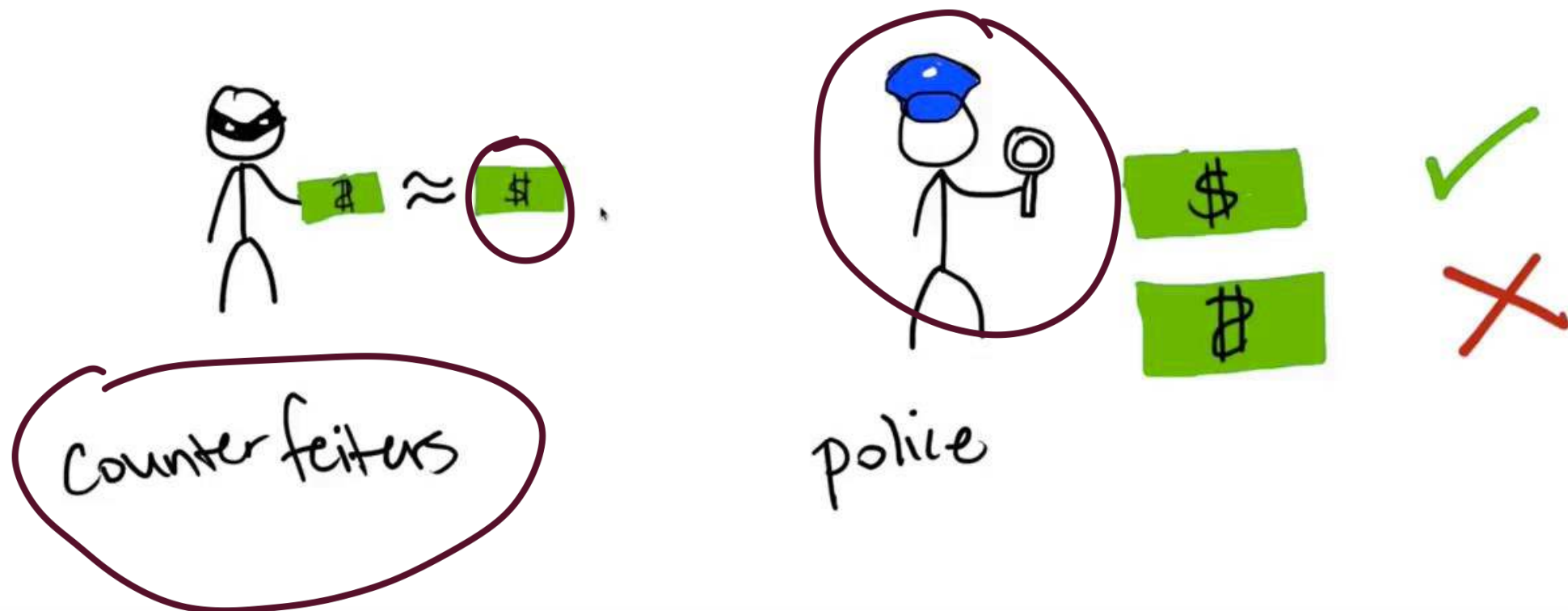
[Shinohara et al. 2017]



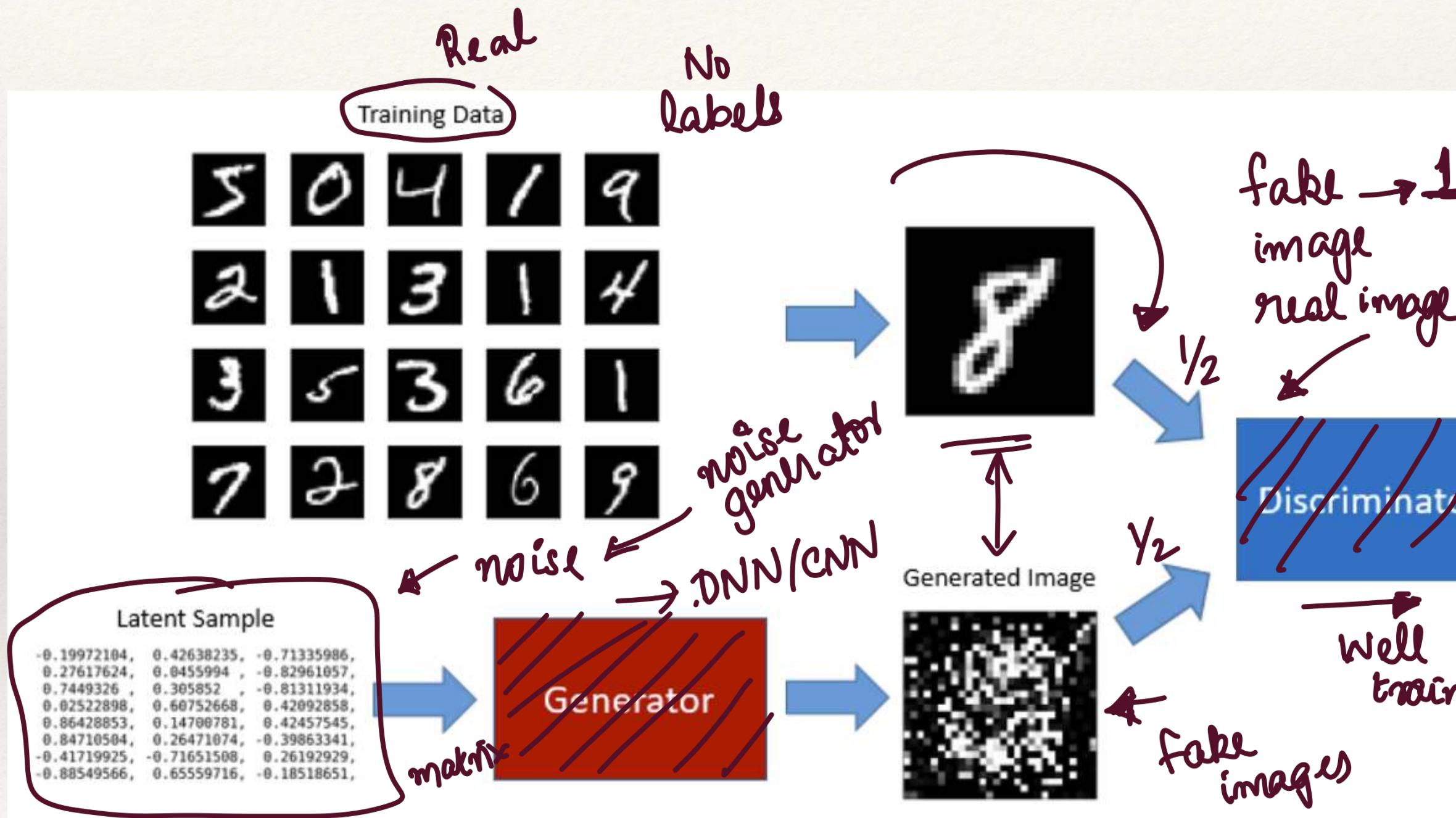
Generative Adversarial Networks

600 x 400

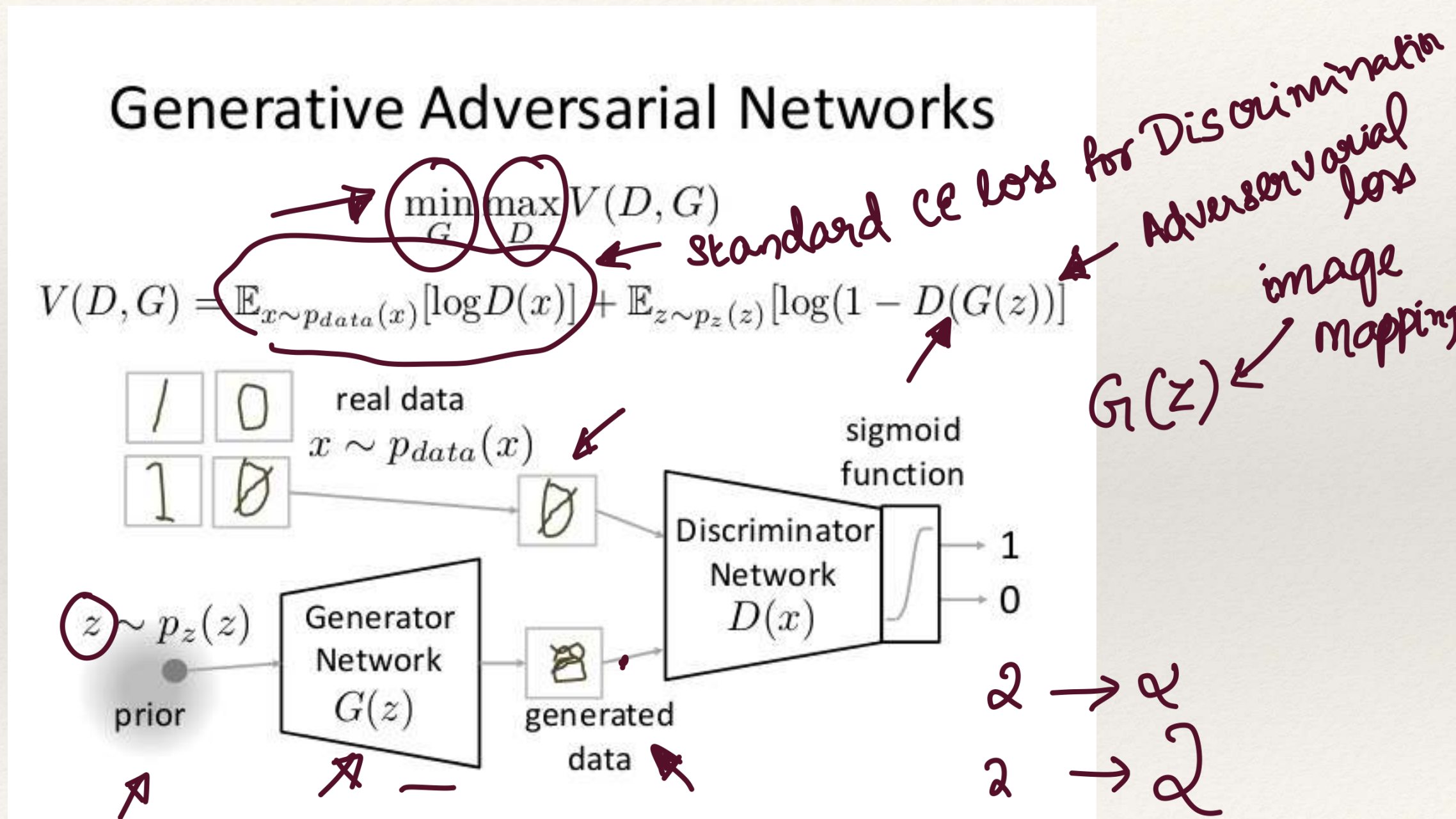
Generative Adversarial Networks (GANs)



Generative Adversarial Networks



Generative Adversarial Networks



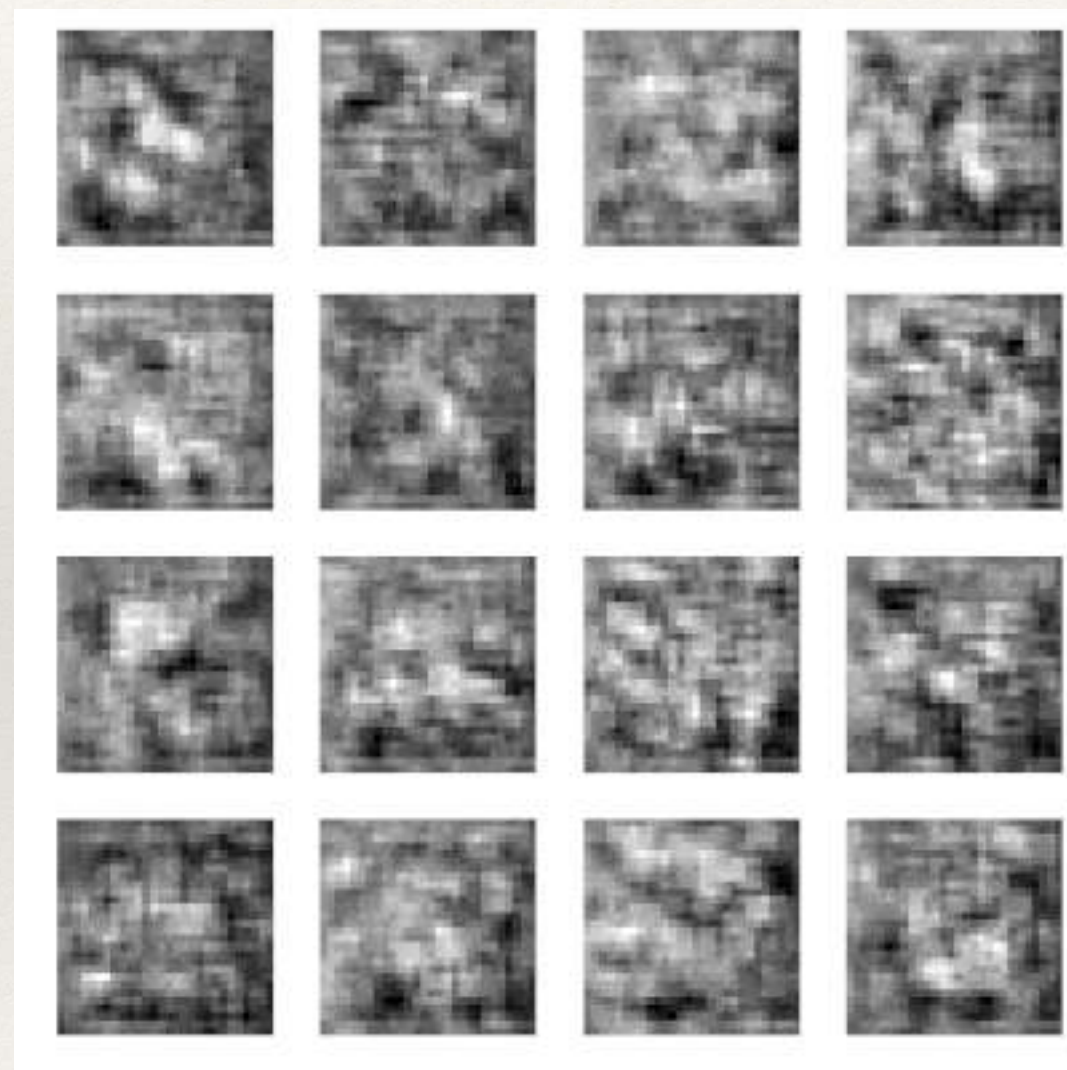
June 4th



usual
class
time

6:00pm

Generative Adversarial Networks



DCGANs